Operation Ember Ascent: Mustafar Flight Report

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PS1 OpenRocket Flight Design Challenge | Body material: Cardboard

1. What we built

We made a two-stage sounding rocket in OpenRocket. It is 60.5 cm long with a 7 cm maximum diameter, and it weighs 2329 g without motors and 3528 g with them, so it is well within the mass requirement of Cardboard. The motors are an I120-15 and an I445-6, impulses being 504 N-s and 476 N-s respectively, total impulse being 980 N-s, comfortably under the 2800 N·s limit. The payload mass is 1kg (1000g).

Recovery. Each stage has its own parachute, and both are set to deploy at apogee with a 0 s delay. Chute diameters: 335 cm and 400cm .

Stability. The brief doesn't set a minimum, so we are just reporting it. At launch the CG is at 28.9 cm and the CP at 37.4 cm (at Mach 0.3), which is 1.2 calibres, or 13.9% of the rocket length.

2. Setting up the Mustafar environment

OpenRocket doesn't have a gravity box in the launch conditions, so gravity is set with the *Planetary Environment (Mustafar)* extension on the simulation, which fixes it at 6.50 m/s². That extension does not touch the atmosphere, so we typed pressure and temperature in by hand. We left “Use International Standard Atmosphere” unticked and let OpenRocket work out the density itself (about 0.373 kg/m³, matching the 0.374 in the brief).

Parameter	Earth (comparison run)	Mustafar (mission)
Gravity	9.81 m/s²	6.50 m/s² (extension)
Pressure	101.3 kPa	75 kPa (750 mbar)
Temperature	15 °C (288 K)	700 K (426.85 °C)
Air density	1.225 kg/m³	0.374 kg/m³
Wind	2m/s average at 90°, 0.2m/s std dev, 10% turbulence	10 m/s average at 270°, 1 m/s std dev, 10% turbulence
Launch site	0 m altitude	0 m altitude, 200 cm rod, 0° angle (vertical)

3. Do we meet the requirements?

Requirement	Our result (Mustafar)	Met?
Apogee 3900 to 4100 m	3913 m	Yes (only just, 13 m above the lower limit)
Payload 0.95 to 1.05 kg	1 kg	Yes
Length at most 220 cm	60.5 cm	Yes
Total impulse at most 2800 N·s	Two I-class motors (at most 1280 N·s combined)	Yes
Deploy at apogee, parachute recovery	Both chutes at apogee, 0 s delay, both stages recovered	Yes
Descent / ground hit velocity 4 to 8 m/s	9.16 m/s	No (see section 5)

4. Mustafar vs Earth

The brief says to treat each environmental effect as a hypothesis and test it by flying the same design in both atmospheres. We flew the exact same rocket (same motors, same chutes) once on Mustafar and once with everything set to Earth. Here is what OpenRocket gave us.

Result	Mustafar	Earth
Apogee	3913 m	2115 m
Time to apogee	33.9 s	20.6 s

Result	Mustafar	Earth
Max velocity	246 m/s (Mach 0.439)	202 m/s (about Mach 0.59, calculated using 340 m/s)
Max acceleration	149 m/s²	145 m/s²
Velocity off the rod	21.8 m/s	22.3 m/s
Velocity at deployment	32.9 m/s	6 m/s
Flight time	1317 s	964 s
Ground hit velocity	9.16 m/s	2.93 m/s
Stability at launch	1.2 cal (at Mach 0.3)	1.2 cal (same, since the geometry is the same)
Downrange drift	11550 m	1735 m

We also did the one-at-a-time runs the brief asks for. Each run below uses Mustafar conditions for everything except the one setting listed, which was switched to its Earth value. Drift is the final lateral distance from the launch pad, exported from OpenRocket.

Run	Apogee	Max vel	Vel at deploy	Ground hit	Flight time	Drift
Mustafar (all Mustafar)	3913 m	246 m/s	32.9 m/s	9.16 m/s	1317 s	11550 m
Earth gravity only	2885 m	231 m/s	33.9 m/s	9.09 m/s	788 s	6749 m
Earth temperature only	2855 m	223 m/s	30.4 m/s	10.1 m/s	1382 s	12510 m
Earth pressure only	3548 m	240 m/s	30 m/s	11.2 m/s	1387 s	12440 m
Earth wind only	4107 m	247 m/s	6.3 m/s	3.82 m/s	1362 s	2404 m
Everything Earth	2115 m	202 m/s	6 m/s	2.93 m/s	964 s	1735 m

In the wind-only run, the wind was set to 2m/s.

The effects of the single changes don't add up to the all-Earth result (-1028, -1058 and -365 m separately, -1798 m together), because drag, gravity and wind act on each other.

Change	Our prediction	What we saw	Did it hold?
Lower gravity (6.5 vs 9.81)	Weight is about 34% lower, so the rocket climbs higher and for longer. Max acceleration should rise by about 3.3 m/s² (just the difference in g).	Putting Earth gravity back drops apogee by 1028 m (3913 to 2885), time to apogee from 33.9 s to 24.9 s, and max acceleration from 149 to 146 m/s², a drop of 3.	Yes. This is the cleanest result. The 3 m/s² change in max acceleration matches the 3.3 we predicted, and gravity alone costs about a quarter of the apogee.
Thinner air (density 0.374 vs 1.225 kg/m³)	Drag scales with density, so less air means less drag, a faster rocket and a higher apogee.	Earth pressure only (denser air, about 1.35 times by the ideal gas law) lowers apogee by 365 m. Earth temperature only (cooler, so denser air, about 2.4 times) lowers it by 1058 m. Everything Earth lowers it by 1798 m.	Yes. More air density gave a lower apogee in every run, and the temperature run, which changes the density more, lost more height than the pressure run. The ideal gas law density figures are our own calculation.
Hotter air (700 K vs 288 K)	Speed of sound goes as the square root of temperature, so it is about 1.56 times higher and the same speed gives a lower Mach number.	Mustafar max Mach 0.439 at 246 m/s, which works out to a sound speed of roughly 560 m/s. On Earth, 202 m/s is about Mach 0.59.	Yes for Mach number. But the temperature run shows that the bigger effect of temperature on our rocket is through air density, not the speed of sound, because we never went near Mach 1.
10 m/s side wind	The rocket weathercocks into the wind, drifts downrange, and arrives at apogee still moving sideways. Ground hit and deployment speeds should include this sideways part.	With the Earth wind setting, drift fell from 11550 m to 2404 m, velocity at deployment from 32.9 to 6.3 m/s, and ground hit velocity from 9.16 to 3.82 m/s. Apogee rose by 194 m.	Yes, and strongly. Wind is the biggest single effect on the descent numbers by far.

Change	Our prediction	What we saw	Did it hold?
Thinner air and lower gravity on the parachute	Terminal speed goes as the square root of (g divided by density), so the same chute should come down faster on Mustafar. More air should slow it down.	Earth gravity only: ground hit 9.09 m/s (no change from 9.16). Denser air (temperature or pressure only): 10.1 and 11.2 m/s, which is higher, not lower.	No. The ground hit number did not follow the terminal speed formula. See section 5.

5. The ground hit velocity: we didn't get it into 4 to 8 m/s

Being straight about this: we did not manage to get the ground hit velocity into the 4 to 8 m/s window, even after trying many times. Our best run has 9.16 m/s, and across all our attempts it sat between about 9 and 12 m/s. We resized both parachutes again and again, and changed the motors to pull the apogee back into range, but nothing brought the number under 8.

The one-at-a-time runs point at the 10 m/s side wind. OpenRocket's ground hit velocity is the total speed, so it includes the sideways speed picked up while drifting under the chutes. With only the wind changed to its Earth setting and the chutes untouched, ground hit velocity fell from 9.16 to 3.82 m/s and velocity at deployment from 32.9 to 6.3 m/s. No other change did that.

A rough check on the exported data agrees. On Mustafar the rocket falls 3913 m in about 1283 s, so it sinks at only about 3 m/s on average, but it drifts 11550 m in that time, about 9 m/s sideways. Together that is about 9.5 m/s, close to the 9.16 m/s OpenRocket reports. These are averages, not measured landing values, so treat them as estimates. The denser-air runs (temperature only and pressure only) landed even faster, at 10.1 and 11.2 m/s. Our guess is that thicker air lets the wind push the rocket more and the flights are longer, but we haven't tested that. So the wind explanation fits our numbers, but we haven't proven it.

With more time we would cut the sideways speed at apogee (longer launch rod, more stability margin) and then re-tune the chutes.

6. Conclusion

On Mustafar our rocket reaches 3913 m, which is inside the 3900 to 4100 m window, and every requirement we can check is met except the ground hit velocity (9.16 m/s against 4 to 8 m/s). Running Earth conditions one at a time, the gravity prediction held clearly (the change in max acceleration matched), the air density prediction held (more air gave a lower apogee in every run), and the temperature prediction held for Mach number, though temperature mattered more through air density. The wind prediction held strongly: wind alone accounts for most of the ground hit velocity, the deployment velocity and the drift. The parachute scaling prediction did not hold, because the ground hit number does not follow the terminal speed formula when there is sideways wind. The biggest thing we learned is that on a windy world the numbers OpenRocket reports can mix up sideways and downward speed, and that you have to look at the vertical part separately before blaming the parachute.